

Design Verification Plan & Report (DVP&R;)

Case ID: VBF-T2R1MIMO-DVPR-01
Date: 2026-08-30
Status: Virtual Verification (Simulation-Based)

Verification Matrix

#	Test	Criterion	Method	Result	Verdict
1	1C discharge energy density	$\geq 327.18 \text{ Wh/kg}$	PyBaMM SPMe + calc-energy	378.69 Wh/kg	PASS ✓
2	4C fast charge (plating)	$V_{\text{anode}} \geq 0$ at all times	PyBaMM SPMe + thermal lumped + plating module	min V = +0.0006 V	PASS ✓
3	4C thermal safety	$T_{\text{max}} \leq 350 \text{ K}$	PyBaMM SPMe thermal lumped	344.95 K	PASS ✓
4	Low-temperature retention	$\geq 90\%$ at -20°C	PyBaMM SPMe lowT_discharge	99.99%	PASS ✓
5	SEI growth @100 cycles	$\leq 500 \text{ nm}$	PyBaMM SPMe aging_1C_100cyc	205.9 nm	PASS ✓
6	SEI growth @500 cycles	$\leq 550 \text{ nm}$	PyBaMM SPMe aging_1C_100cyc (500 cyc)	518.1 nm	PASS ✓

Notes

- All tests are simulation-based (PyBaMM SPMe model with Chen2020 parameterization)
- Electrolyte transport and SEI parameters are overrides (estimated values, not experimentally validated)
- Architecture parameters are design choices within the Chen2020 framework
- Physical prototype testing required for production validation